Matrix Multiplication Accelerator

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ECE 520
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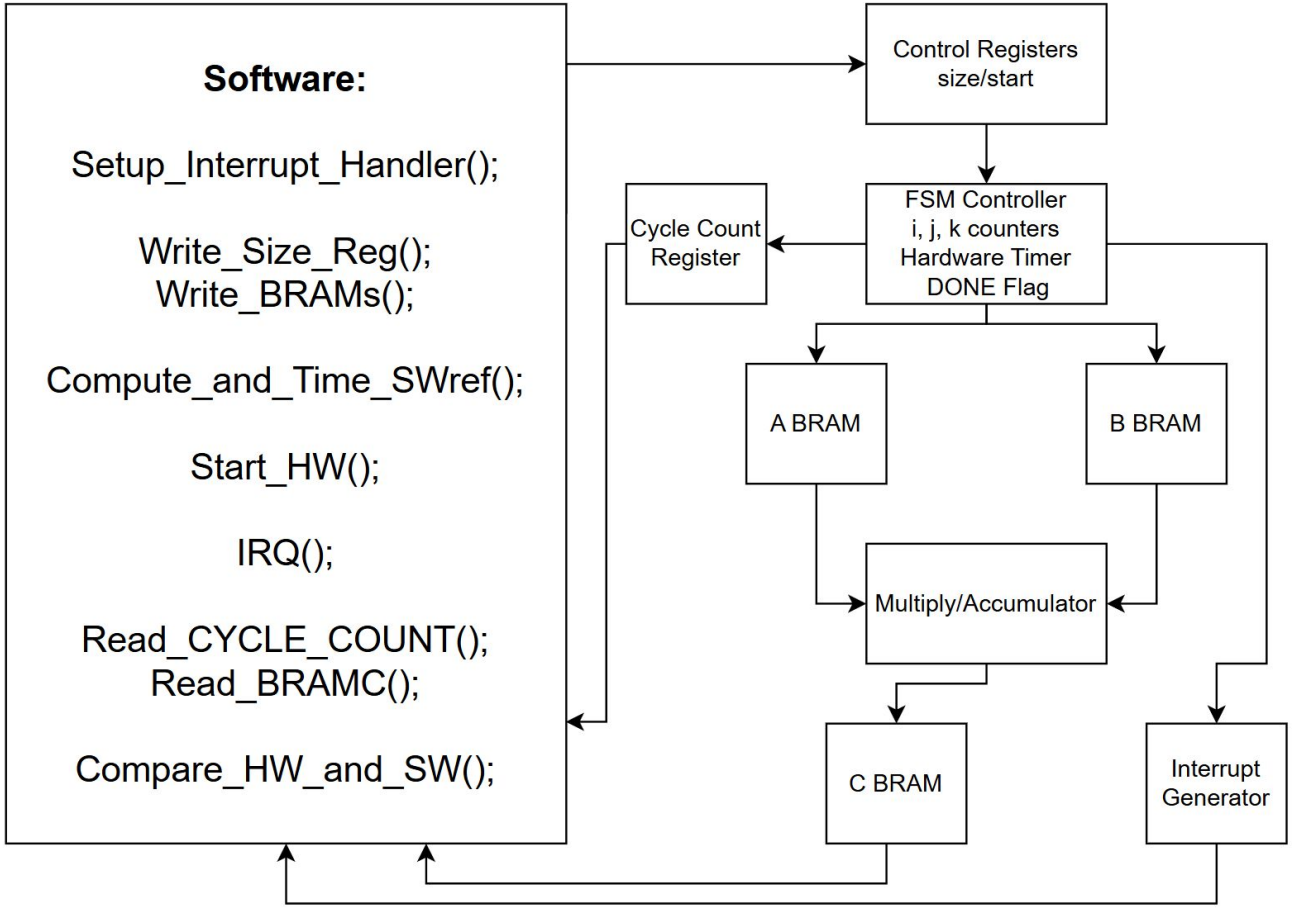
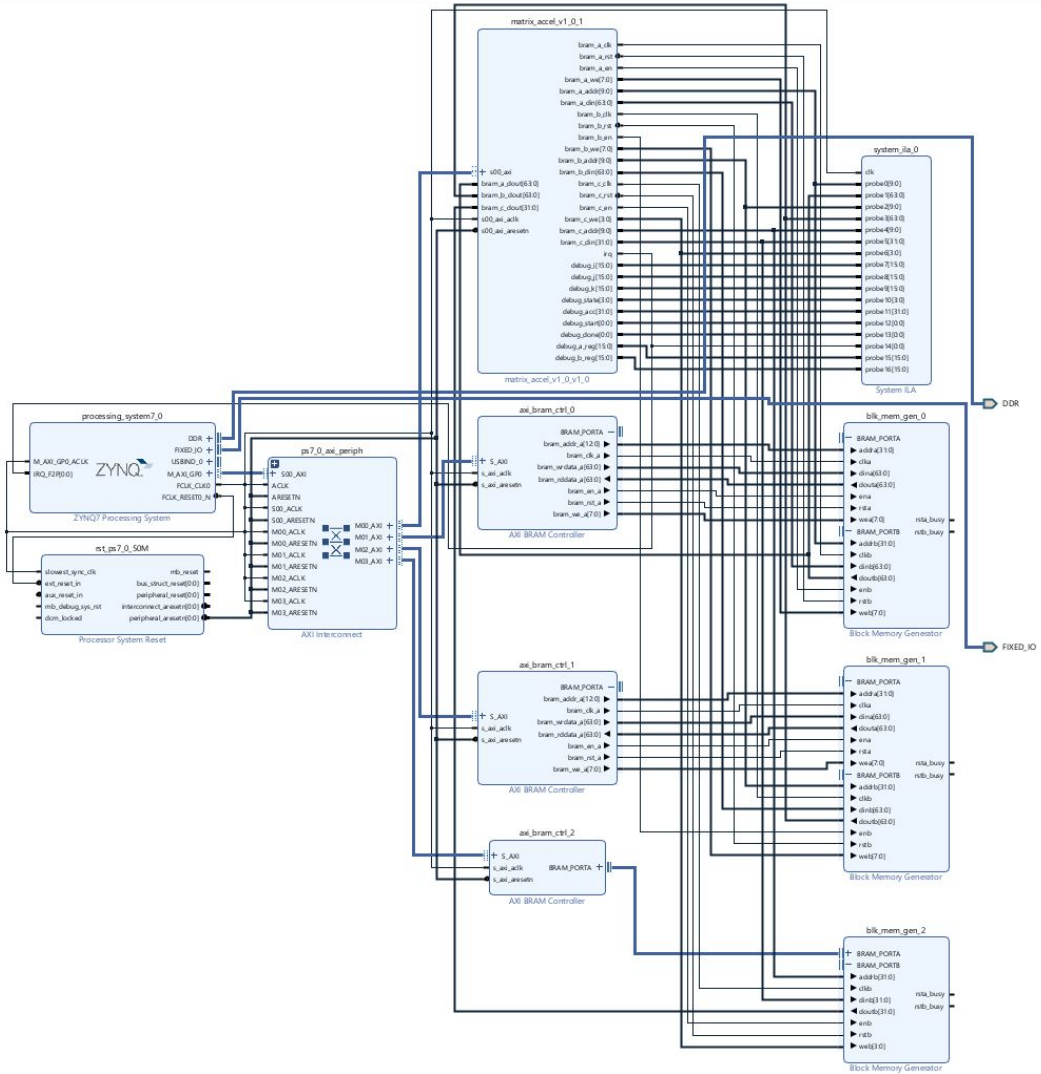
System Overview

- Employs matrix operation: A x B = C
- PS configures AXI-Lite registers (size, int_en, reset)
- PS writes Matrices A/B to BRAM
- PS computes a software reference for Matrix C
- PS writes the START_bit_mask to AXI-Lite Control register to start the Accelerator
- PL Hardware execution using MAC architecture
- PL writes the Matrix C result to BRAM
- PS reads from BRAM C and compares HW and SW results

Block Diagram



Final Vivado Block Design



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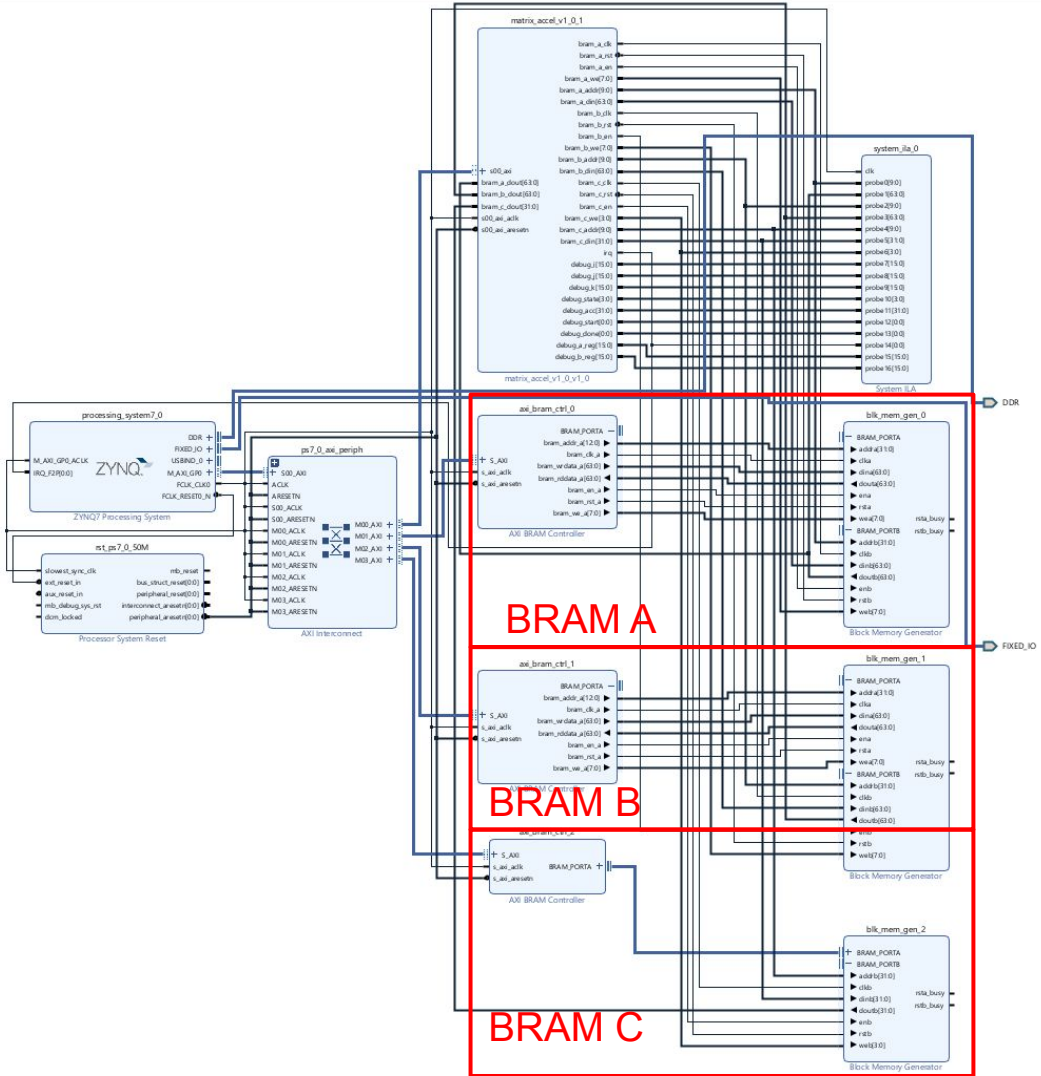
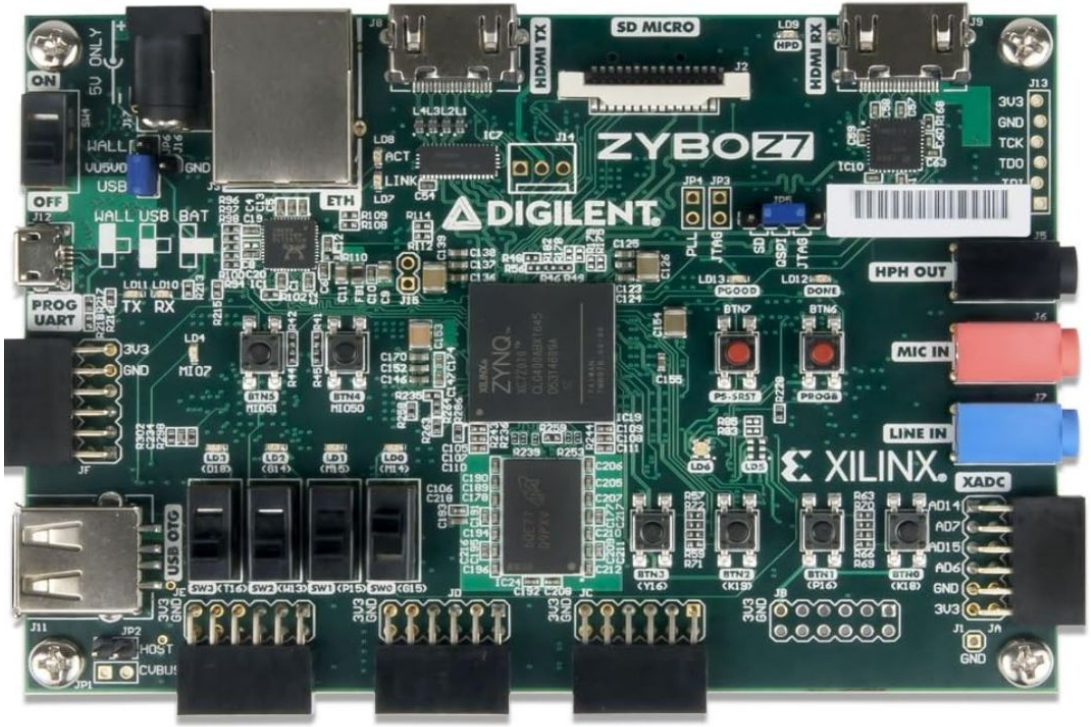


Components Used

- Zybo Z7 ARM/FPGA SoC Board

- USB-A to USB Micro-B Cable



AXI-Lite Control Registers

Offset	Name	Description
0x00	CONTROL	Start, interrupt enable, soft reset
0x04	SIZE	Matrix dimension (N)
0x08	CYCLE_COUNT	Execution cycle count
0x0C	STATUS	Busy/done flags, FSM state

Matrix Multiplication Conceptual Background

[row][column]

$$\boxed{C[i][j] = \sum_{k=0}^{N-1} (A[i][k] * B[k][j])}$$

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} \cdot \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$$

$$= \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \\ a_{31}b_{11} + a_{32}b_{21} & a_{31}b_{12} + a_{32}b_{22} \end{bmatrix}$$

BRAM Address Generation (1-D)

1. Address_A_Index = i*N + k
2. Address_B_Index = j*N + k
3. Address_C_Index = i*N + j

Where:

i = row index

j = column index

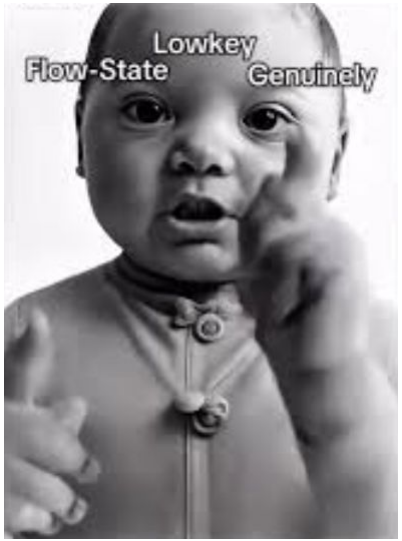
N = Size of matrix

Matrix Calculation

- Once initial BRAM A/B addresses are calculated, elements are latched
  - A_reg and B_reg
- Accumulator is incremented by A_reg * B_reg
- New BRAM A/B addresses are calculated (k increments)
- **Repeat until k = N-1**
- Accumulator now holds the final value for C[0][0]
- Write result to BRAM C and increment Matrix C's column index (j)
- **Repeat until Matrix C's first row is filled (j = N-1)**
- Increment Matrix C's row index (i)
- **Repeat until Matrix C is full (i = N-1)**

State Selection/Sequential Logic

1. ST_IDLE (default) - waits for the PS to write to the START register
2. ST_CLEAR_ACC
3. ST_READ_AB (present addresses)
4. ST_WAIT_AB1 (1-cycle read latency)
5. ST_WAIT_AB2 (1-cycle settling time)
6. ST_LATCH_AB
7. ST_MAC
8. ST_CHECK_K - returns to ST_READ_AB if (k < N-1)
9. ST_WRITE_C
10. ST_CHECK_J - returns to ST_CLEAR_ACC if (j < N-1)
11. ST_CHECK_I - returns to ST_CLEAR_ACC if (i < N-1)
12. ST_DONE



Block Memory Generator Interface

Signal Name	Description	Custom IP Implementation
clk	Clock signal	Tied to PS's axi_aclk
rst	Reset signal	Tied to 0; not used. The BRAM units always hold the last written data
en	Enable signal for BRAM access	Asserted when the accelerator performs a read or a write; deasserted when idle
we	Write enable	Determines read vs write; '0' = read, '1' = write
addr	Address bus	Specifies which memory location to access
din	Data input bus	Carries data from accelerator
dout	Data output bus	Carries data to accelerator

Software: Menu User Interface

1. Load default matrices into software buffers.
2. Write matrices A and B into BRAM.
3. Compute the software reference (timed).
4. Run the hardware accelerator using interrupts.
5. Read matrix C from BRAM and compare HW/SW results.
6. Print accelerator registers.
7. Print matrices A, B, C_hw, and C_sw.
8. Enable interrupts.
9. Disable interrupts.
10. Run the full matrix test using interrupts (1-5).
11. Assert the hardware accelerator’s soft reset.
12. Change the matrices’ sizes.
13. Print hardware control registers’ base addresses and interrupt information.
14. Display this menu.

Throughput Scaling via Parallelism

- SW and HW calculation methods are equivalent; how do we leverage PL resources to increase throughput?

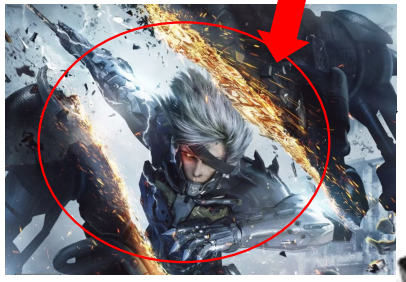
Increasing RTL complexity is easy; we need to send more data at once!

- **Each Matrix A/B element is an int16**
- 1-MAC: 16-bit data bus
- 2-MAC: 32-bit data bus
- 4-MAC: 64-bit data bus
- **Each Matrix C element is an int32**

Throughput Scaling via Parallelism cont. (4-MAC)

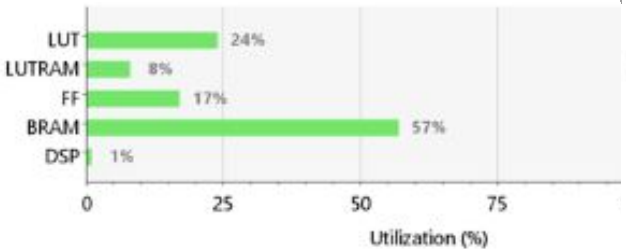
- Need a packing helper function in SW to pack 4 int16s into an int64

- Need bit-slicing registers in HW to extract 4 int16s from an int64

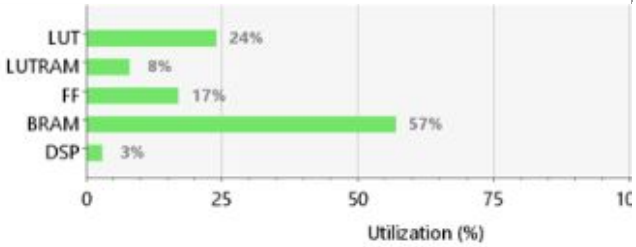


Results: Resource Utilization

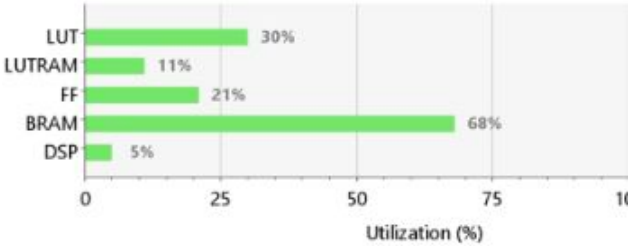
1-MAC



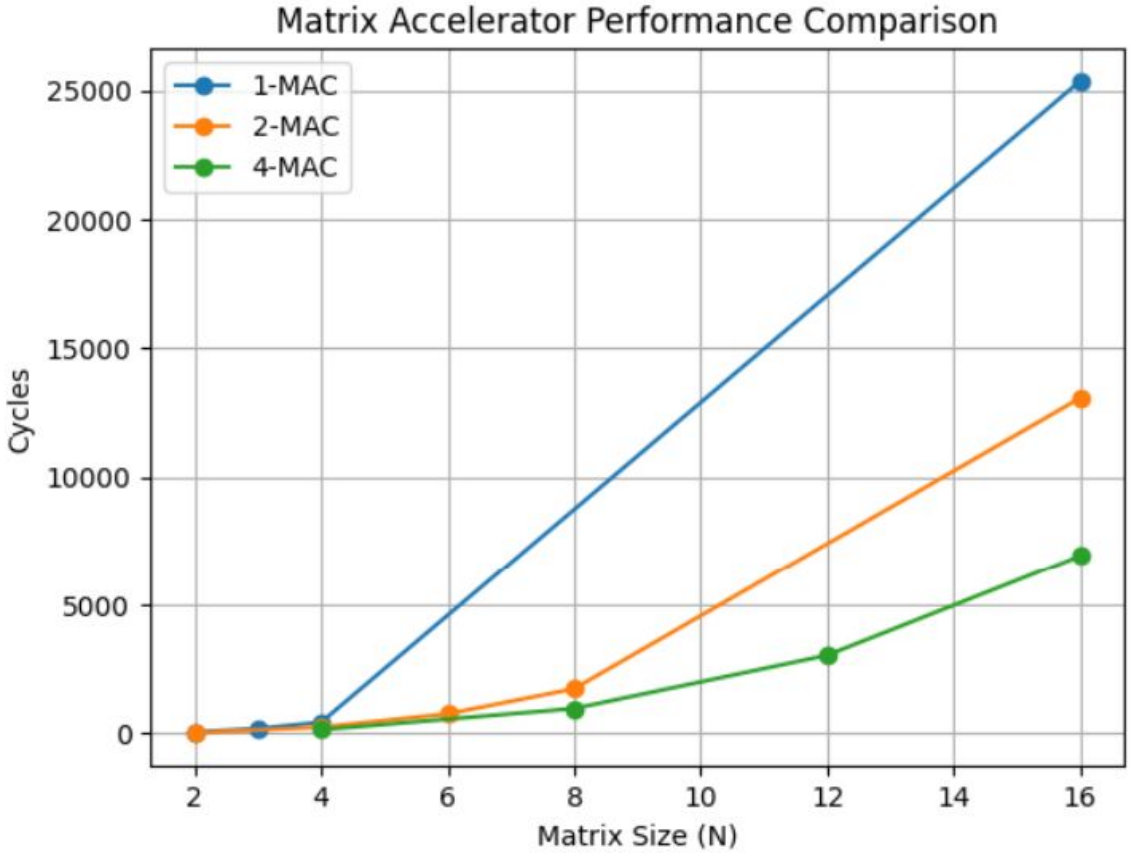
2-MAC



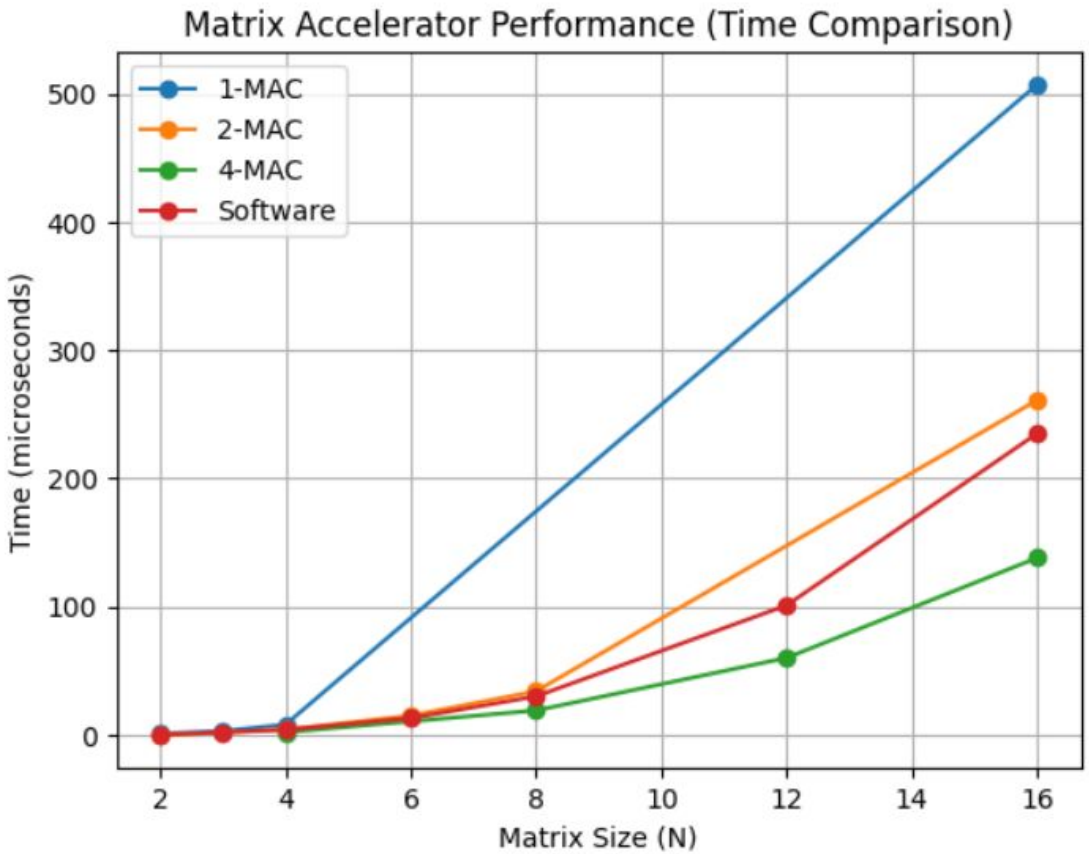
4-MAC



Results: Performance Scaling



Results: HW vs SW Compute Time



Demonstration Video

https://youtu.be/MQN0aFwF9Ak

AI Tool Usage

- Hardware:
  - Guide implementations
  - Write testbenches for matrix accelerator IP prototypes
  - Debug incorrect functionality and IP updating issues
- Software (functions):
  - Enhance user input by allowing multiple input characters via UART buffer helper function
  - Scale BRAM write functions to 2-MAC and 4-MAC

- Increasing productivity without decreasing conceptual understanding!

Conclusion

This project explored:

- AXI4 handshaking interfaces
- PS-PL communication techniques
- PL-based parallelism scaling
- BRAM-based memory architecture
- Finite state machine control design
- AXI-Lite register designation
- Hardware-software co-design for performance optimization and verification

References

[1] AMD Xilinx, *AXI Reference Guide*, UG1037.

[2] AMD Xilinx, *Zynq-7000 SoC Technical Reference Manual*, UG585.

[3] AMD Xilinx, *Block Memory Generator v8.4 Product Guide*, PG058.

[4] AMD Xilinx, *AXI BRAM Controller v4 Product Guide*, PG078.

[5] AMD Xilinx, *Vivado Design Suite User Guide: Designing with IP*, UG896.

References (Other)

- https://collider.com/new-matrix-movie-warner-bros-mike-de-luca-pam-abdy/
- https://www.amazon.com/Digilent-Zybo-Z7-Zynq-7000-Development/dp/B0797G4485?th=1
- https://www.youtube.com/watch?v=u6MVJiOw2d4
- https://www.clipartmax.com/middle/m2H7H7N4G6Z5b1b1_related-image-thumb-up-emoji/
- https://www.pinterest.com/pin/body-reference--295267319341675447/
- https://tenor.com/view/adress-me-elephant-in-the-room-gif-351933056326669780
- https://www.instagram.com/reel/DSIa7MAEeFg/
- https://www.reddit.com/r/REPOgame/comments/1kmonk1/low_quality_hideo_kojima_thinking_room/#lightbox
- https://www.polygon.com/2012/12/16/3775302/metal-gear-rising-revengeance-north-america-box-art-highlights/
- https://royal.com/product/royal-6000ml-electronic-cash-register/
- https://laurapickens.wordpress.com/2014/11/18/homework/
- https://www.ubisoft.com/en-us/game/rainbow-six/extraction
- https://usagif.com/gif/confetti-31/